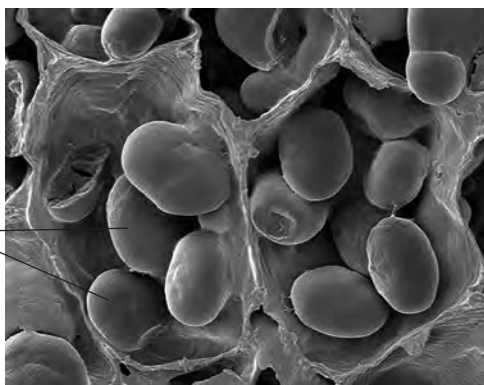


- 6 The cassava root is an important food crop grown in the tropics.

The photograph shows organelles containing starch granules inside cassava root cells, as seen using an electron microscope.

organelles
containing starch
granules



(Source: © Science Photo Library / Alamy Stock Photo)

- (a) (i) Name these organelles found in cassava root cells.

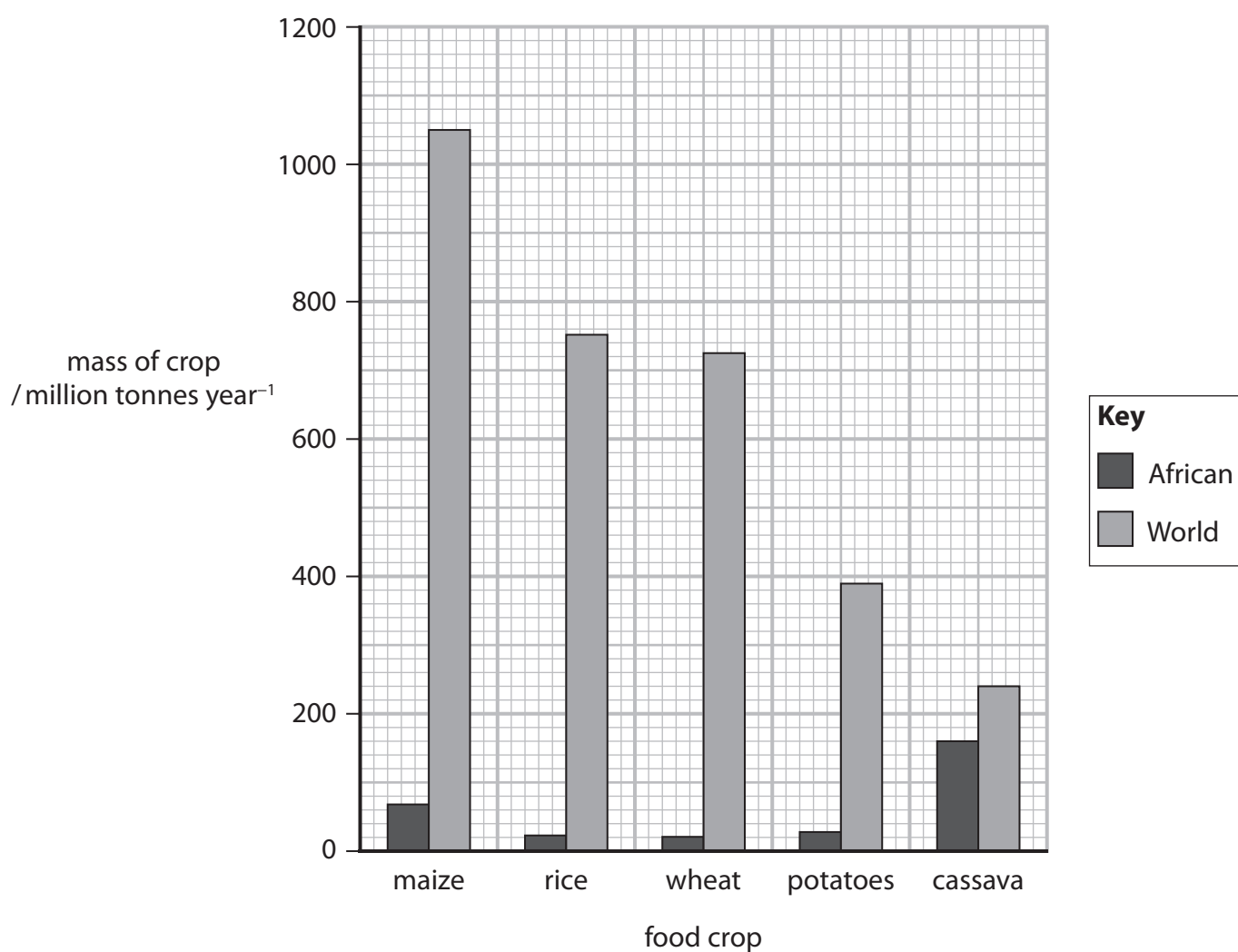
(1)

- (ii) Explain how the structure of starch relates to its function.

(3)



- (b) The graph shows the mass of five food crops grown in Africa. It also shows the world production of these crops.



- (i) Which is the percentage of the world cassava crop that is grown in Africa?

(1)

- ☐ **A** 33 %
- ☐ **B** 34 %
- ☐ **C** 66 %
- ☐ **D** 67 %



- (ii) It is predicted that the world maize production will increase by 2.3 % in the following year.

Calculate the world production of maize in the following year.

Give your answer to the nearest whole number.

(1)

Answer million tonnes

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P 6 7 1 0 3 A 0 1 7 2 8

*(c) Some plants contain chemicals that protect them from being eaten by animals.

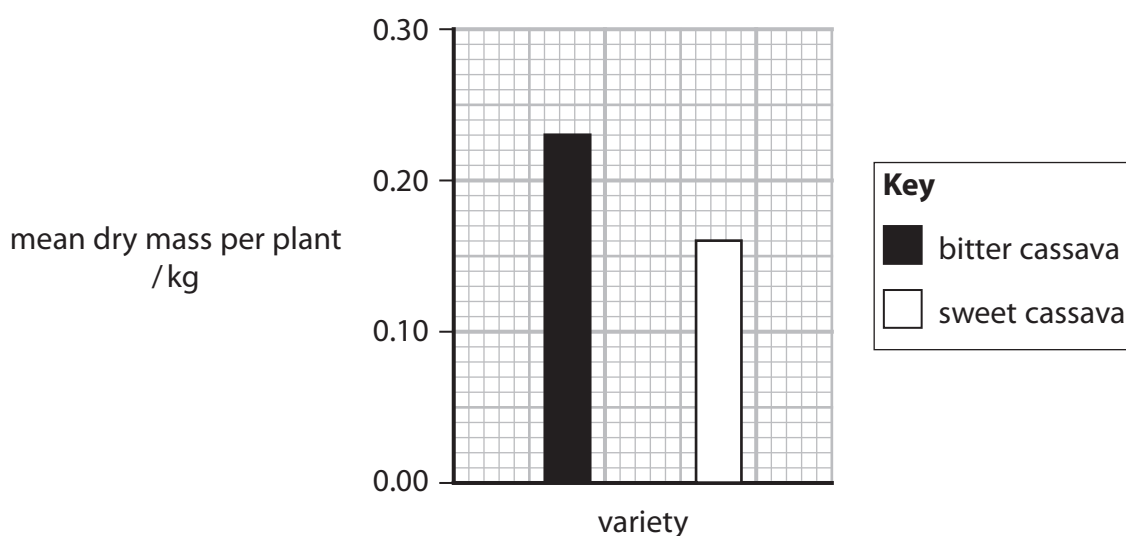
These chemicals taste bitter and can be converted by enzymes into hydrogen cyanide.

Hydrogen cyanide inhibits respiration and can cause death.

Cassava can be grouped into two main varieties: bitter cassava and sweet cassava. These varieties differ in the concentration of the bitter-tasting chemicals.

Processing of the cassava removes most of these chemicals.

The graph shows the mean dry mass of these cassava varieties when grown in the same conditions.



The table gives information about three crop plants.

Crop plant	Water requirements	Minimum level of nutrients required in soil	Typical mass of bitter tasting chemicals / mg kg^{-1}	Typical mass of carbohydrate / 100 g of dry plant matter	Typical mass of protein / 100 g of dry plant matter
Bitter cassava	low to medium	low	100 to 500	94.70	1.80
Sweet cassava	low to medium	low	10 to 50	94.62	1.84
Maize	medium to high	medium to high	1 to 2	77.46	8.75



Discuss the advantages and disadvantages of growing these three crop plants for human food.

Use the information in the question to support your answer.

(6)

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(Total for Question 6 = 12 marks)



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